

Bridging the Digital Divide: A Technical and Financial Analysis of the Community-Based Broadband Deployment in the Tusheti Region

1. Abstract

The Tusheti region, situated on the northern slopes of the Greater Caucasus, represents one of the most geographically isolated areas in Georgia. Historically, the digital divide in this region has been exacerbated by extreme terrain and low population density, rendering traditional fiber-optic infrastructure economically and logistically unfeasible. This paper analyzes a strategic intervention led by the Internet Society (ISOC) Georgia Chapter and the Tusheti Development Fund (TDF) to deploy a sustainable, community-based fixed wireless broadband network. Utilizing Ubiquiti and MikroTik technologies, the project transitioned from standard low-altitude planning to a specialized high-altitude network topology. This shift was necessitated by the "twisted relief" of the Caucasian ridges, where the high density of evergreen foliage and significant winter snow accumulation—averaging 1.5 meters—threatened Fresnel Zone clearance and overall link stability.

The technical architecture is supported by an off-grid solar energy model, which was scaled from 12 to 20 panels to compensate for reduced charging efficiency during winter months, where a helpful solar azimuth of 53° was physically impossible to maintain on high-elevation ridges. Beyond structural engineering, the project addressed unique geological challenges, specifically high sulfide and ferrite mineral concentrations that attract frequent lightning strikes, by implementing Andeli surge protection systems. The implementation successfully connected approximately 60 consumers, including critical tourism hubs, the Border Guard Authority, and the Protected Areas Agency. This study demonstrates that adapting hardware for extreme environmental resilience, coupled with a community-led non-profit financial model, provides a scalable blueprint for rural connectivity in mountainous terrains.

2. Introduction

Tusheti is located in northeast Georgia, bordering the Russian autonomous republics of Chechnya and Dagestan. Its high-altitude geography and lack of year-round road access have traditionally isolated its communities from the national digital infrastructure. The primary objective of the Tusheti connectivity project was to establish a resilient and sustainable fixed wireless network to support the local tourism-based economy, which demands an aggregate bandwidth of 230–270 Mb/s, alongside essential education and healthcare services.

As outlined in the "Timeline - Project Implementation Plan," the project pursued five core operational goals:

1. Procurement and installation of specialized wireless equipment as specified in the consolidated project budget.

2. Implementation of technical training sessions to enhance the digital and maintenance skills of local staff.
3. Adherence to a deployment rate of one access point installation per day during the active implementation phase.
4. Integration of at least twelve prominent hotels and guest houses into the network.
5. Specialized training for two local IT specialists in the Tusheti district, focusing on LAN wiring, indoor switching, IP configuration, and troubleshooting.

3. Organizational Framework and Legal Governance

The project was facilitated through a multi-stakeholder partnership formalized in a Memorandum of Understanding (MoU) signed on August 1, 2016. This collaborative framework ensured that technical expertise, financial resources, and local management were aligned. Legal and administrative support was further bolstered by the Tbilisi IOT Lab (Tax ID: 404527320), providing a formalized registry for technical oversight.

Table 1: Stakeholder Responsibility Matrix

Stakeholder	Primary Responsibilities
Internet Society (ISOC)	Provision of primary grant funding and international project oversight.
ISOC Georgia Chapter / IDI	Technical/marketing training, tariffication strategy, and a three-year monitoring mandate.
Tusheti Development Fund (TDF)	Local operation, maintenance, client installation, and ownership of physical network assets.
Small & Medium Telecom Operators Association / LTD Freenet	Provision of 100 Mb/s free transit at Ruispiri; technical engineering and hardware procurement.
Technical Experts (K. Stalinsky, M. Charelidze)	Network design, link engineering, and comprehensive budgetary oversight.

4. Technical Methodology and Network Topology

4.1. Network Architecture and Site Selection The original scheme proposed by the Georgian National Communications Commission (GNCC) suggested placing masts at lower altitudes near villages. Field assessments revealed this approach was flawed due to "twisted relief" and Line of Sight (LOS) obstructions. Because households and hotels are often 200m apart on steep slopes, a village-level mast cannot provide LOS to all. Consequently, a "high altitude" topology was adopted. By placing masts on higher ridges, the project ensured direct Fresnel Zone clearance and extended coverage to remote areas like Tsova-Tusheti and the village of Indurta.

4.2. Equipment Specifications The network utilizes high-performance wireless hardware designed for extreme Point-to-Point (PtP) and Point-to-MultiPoint (PtMP) resilience:

- **Backhaul Links:** Ubiquiti AF-5x (5GHz) paired with RD-5G34 34dBi dishes and RAD-3RD RocketDish Radomes for atmospheric protection.
- **Sectoral Antennas:** Primarily the LBE-5AC-16-120 and AM-5G19-120 models, offering 90° to 240° coverage to maximize the consumer footprint per mast.
- **Routing and POE Management:** MikroTik RB750P-PBr2 (PowerBox) and RB2011UiAS-IN for distribution and Power over Ethernet (POE) delivery.
- **Client CPE:** MikroTik SXT-5 and LiteBeam M5 units were deployed at individual subscriber locations.

4.3. Link Engineering The backbone originates in the Alazani Valley at Ruispiri, extending to Abano Pass, and then to the primary distribution hub at Diklo Mountain. From Diklo, the network branches into critical distribution vectors:

1. Diklo to Dochu and Koklata.
2. Diklo to Dano and Chigho.
3. Diklo to Omalo (serving both Upper and Lower Omalo).
4. Specific high-altitude links to Jvarboseli to ensure deep gorge penetration.

5. Environmental Adaptation and Resilience

5.1. Weather-Resistant Mast Engineering Masts were engineered to survive extreme "snow-slip" pressures and high-velocity winds. The structures utilize steel tubes with a 76mm outer diameter and 3mm wall thickness, manufactured to ISO 9001:2008 standards. A critical technical detail involves the use of 82mm x 3mm sleeves for joining sections to ensure structural rigidity. These were secured using M300 grade concrete foundations (1/8 cubic meter per site) and specialized anchors for rocky terrain.

5.2. Lightning and Static Mitigation The geology of Tusheti—characterized by high sulfide and ferrite metal content—renders equipment highly susceptible to electrical surges. Previous commercial attempts by Mobitel (Beeline) resulted in thousands of dollars in equipment losses due to burn-out. To mitigate this, the project implemented specialized Andeli thunder protection systems and dedicated galvanic grounding to provide necessary isolation.

5.3. Winter Survival Strategies To maintain operational integrity during heavy snowfall, mast heights were set at a minimum of 4 meters. This ensures that radio equipment remains clear of the 1.5-meter snowdrifts and prevents damage from moving snow masses on steep slopes.

6. Sustainable Energy Solutions

An off-grid solar solution was required to power the infrastructure. The system uses a 24V configuration consisting of 100W/12V solar panels feeding a POE-BOX8-G distribution system.

Efficiency Limitations and Adaptation: The initial requirement of 12 panels was increased to 20 (4 panels per major mast site). Engineering analysis determined that the winter solar azimuth on the ridges made a "helpful angle" of 53° impossible to maintain. This constraint, combined with the reduced efficiency of poly-crystal battery charging during overcast periods, necessitated the increase to 200W/12V combined power per mast. This ensures the 100a/h batteries remain charged through low-irradiance cycles.

7. Financial Analysis and Consolidated Budget

Procurement was optimized through a fixed 5.4%–5.6% discount secured through the partnership with LTD Freenet and Intelcom Group. The current topology was specifically engineered with extra capacity to facilitate future expansion into the Khevsureti region.

Table 2: Consolidated Project Financial Breakdown (\$USD)

Category	Item Description	Sub-Total
Radio Equipment	AF-5x links, RD-34 dishes, Radomes, PBE-5AC-620s, LBE-5AC sectors, Rocket M5s, and MikroTik routers/injectors.	\$11,282.00
Energy Systems	20 Solar panels, 100a/h high-capacity batteries, and Nippotec charge controllers.	\$5,695.00
Infrastructure	Steel masts (76mm/82mm sleeves), M300 concrete, fasteners, anchors, and outdoor rack mounts.	\$6,373.00
Operational & Logistics	Fuel (non-plastic cans), Delica L300 rental, cargo transport, and GuestHouse accommodations (Omalo/Diklo).	\$4,541.20
Human Resources	Honorariums for engineers (TCP/IP, Tower Designer), Directors, and Operators (including 20% income tax).	\$11,200.00

Total Project Budget		\$39,091.20
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8. Implementation Results and Community Impact

The deployment successfully bridged the connectivity gap, serving approximately 60 consumers. Beneficiaries include prominent hospitality providers such as Hotel Shina and Samtsikhe (which utilizes MikroTik SXT-5 hardware), the Border Guard Authority, and the Protected Areas Agency.

Service and Sustainability Model: The TDF operates as a non-profit entity ensuring regional sustainability:

- **Profit Reinvestment:** Profits are capped at 10% above OPEX/CAPEX, with all surplus funds mandated for network upgrades.
- **Tariff Structure:** To ensure local affordability, the installation fee is set at 100–120 GEL (covering the CPE), with monthly subscriptions ranging from 25–30 GEL.

9. Conclusion

The Tusheti project serves as a definitive model for high-mountain rural broadband. By prioritizing high-altitude topology to solve the "twisted relief" problem and implementing robust solar and lightning defenses, the network has achieved a level of resilience that commercial operators could not sustain. The success in Indurta and Tsova-Tusheti confirms the scalability of this community-led framework, which is currently being applied to the expansion into the Khevsureti region.

10. References

1. Memorandum of Understanding (MoU): Tusheti Internet Provision Project (August 1, 2016).
2. Tusheti Development Fund - ISOC Interim Implementation Report V2.
3. Technical Tower Drawings and Specifications (V. Dvalishvili, Drawing #599-357-991).
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5. Tbilisi, Registry (Tax ID: 404527320).
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